Name- Shreyash Khapekar
PRN - B25ET1019
Division- SY1 ENTC

Write a program in C++ to implement string class. Write constructors, destructors, Accepts function and Display function.

```
#include <iostream>
#include <cstring>
using namespace std;

class String {
    char *str;
    int len;

public:
    String() {
        len = 0;
        str = new char[1];
        str[0] = '\0';
    }

    String(const char *s) {
        len = strlen(s);
        str = new char[len + 1];
        strcpy(str, s);
    }

    ~String(){
        delete[] str;
    }

    void accept() {
        char temp[100];
        cout << "Enter a string: ";
```

```
        cin >> temp;

        delete[] str;
        len = strlen(temp);
        str = new char[len + 1];
        strcpy(str, temp);
    }

    void display(){
        cout << "String: " << str << endl;
    }
};

int main() {
    String s1;
    cout << "Testing default constructor:\n";
    s1.display();

    String s2("Hello");
    cout << "\nTesting parameterized constructor:\n";
    s2.display();

    cout << "\nTesting accept function:\n";
    s1.accept();
    cout << "Entered string is:\n";
    s1.display();

    return 0;
}
```